

Kickstart Linux Kernel Contribution by Fixing Coverity Issue

Robertus Diawan Chris

Who am I?

- Used to work as web developer (mostly backend)
- On career break
- Sometimes doing programming live stream on youtube
- Resources available on
github.com/bruhtus/opensuse-asia-summit-2026

Where do we start?



KernelNewbies : KernelJanitors

Last updated at 2017-12-30 01:29:53

From the [Kernel Janitors website](#) : «We go through the linux kernel sources, doing code reviews, fixing up unmaintained code and doing other cleanups and API conversion. It is a good start to kernel hacking.»

If the site is down, there's a local copy of all the content at: <http://kernelnewbies.org/MigratingInProgress>

"The Kernel Janitors Project" is a good point to start for newbie kernel hackers. The project maintains a [TODO list](#) from which contributors can pick a job which fits their skills. The tasks are mainly code cleanups so one can dive into kernel code without knowing all the technical details already. It's also a good point to learn about how to (or how not to) submit patches to the kernel.

The project keeps its own mailing list (<http://vger.kernel.org/vger-lists.html#kernel-janitors>) and the maintainer (Alexey Dobriyan at the moment) also irregularly produces a patchset which can be found on <http://coderoack.org/kj/>.

Core subsystems

- [Core API Documentation](#)
- [Driver implementer's API guide](#)
- [Memory Management Documentation](#)
- [Power Management](#)
- [Scheduler](#)
- [Timers](#)
- [Locking](#)

Human interfaces

- [Input Documentation](#)
- [Human Interface Devices \(HID\)](#)
- [Sound Subsystem Documentation](#)
- [GPU Driver Developer's Guide](#)
- [Frame Buffer](#)
- [LEDs](#)

Networking interfaces

- [Networking](#)
- [NetLabel](#)
- [InfiniBand](#)
- [MHI](#)

Storage interfaces

- [Filesystems in the Linux kernel](#)
- [Block](#)
- [CD-ROM](#)
- [SCSI Subsystem](#)
- [TCM Virtual Device](#)
- [NVMe Subsystem](#)

Other subsystems

Fixme: much more organizational work is needed here.

- [Accounting](#)
- [CPUFreq - CPU frequency and voltage scaling code in the Linux\(TM\) kernel](#)
- [EDAC Subsystem](#)
- [FPGA](#)
- [I2C/SMBus Subsystem](#)
- [Industrial I/O](#)
- [PCMCIA](#)
- [Serial Peripheral Interface \(SPI\)](#)
- [1-Wire Subsystem](#)
- [Watchdog Support](#)
- [Virtualization Support](#)
- [Hardware Monitoring](#)
- [Compute Accelerators](#)
- [Security Documentation](#)
- [Crypto API](#)
- [BPF Documentation](#)
- [USB support](#)
- [PCI Bus Subsystem](#)
- [Assorted Miscellaneous Devices Documentation](#)
- [PECI Subsystem](#)
- [WMI Subsystem](#)
- [TEE Subsystem](#)

```
>> - [WmiDataId(4), read, Description('Minumum Fan Speed.'),  
>> WmiSizeIs("NumOfFans")] uint32 FanMinSpeed[];  
>> + [WmiDataId(4), read, Description('Minimum Fan Speed.'),  
>> WmiSizeIs("NumOfFans")] uint32 FanMinSpeed[];  
>> };  
>> --  
>> 2.53.0  
>Looks good.  
>Reviewed-by: Mark Pearson <mpearson-lenovo@squebb.ca>
```

Hi gents,

The MOF spelling mistakes are well known. We've left them is as to ensure match with what the hardware actually reports.

See: <https://lore.kernel.org/platform-driver-x86/cfd7977e-d612-4e08-a68a-65fed8e164b6@gmx.de>

I suppose if we're going to continue getting these types or PR I should add a note to the documentation. I'll add that soon.

Thanks,
Derek

COVERITY / LINUX KERNEL BUGS / MENTORING / PRESENTATIONS

ONE simple and rewarding way to contribute to the Linux Kernel: Fix Coverity issues

Posted on September 28, 2024 by Gustavo Corra Silva / 0 Comment



Analysis Tool Type

- Static Analysis (before the program run)
- Dynamic Analysis (when the program run)

Example of Analysis Tool

- Coverity Scan (static analysis)
- Sparse (static analysis)
- Coccinelle (static analysis)
- Smatch (static analysis)
- Syzkaller (dynamic analysis)

<https://scan.coverity.com/>

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COVERITY SCAN STATIC ANALYSIS

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- ✓ The root cause of each defect is clearly explained, making it easy to fix bugs
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Apache Hbase fixed 75% of Resource leak defects found by Coverity Scan. [Read more >>](#)



Coverity Scan: Linux

Project Name	Linux
Lines of code analyzed	23,694,413
On Coverity Scan since	Feb 24, 2006
Last build analyzed	10 months ago
Language	C/C++
Repository URL	https://git.kernel.org/
Homepage URL	https://kernel.org/
License	GPL (GNU General Public License version)

Want to view defects or help fix defects?

 Add me to project

Analysis Metrics

Version: v6.18

Dec 01, 2025

Last Analyzed

23,694,413

Lines of Code Analyzed

0.83

Defect Density

Defect changes since previous build dated Nov 24, 2025 

Coverity Scan: linux-next weekly scan

Project Name	linux-next weekly scan
Lines of code analyzed	34,355,053
On Coverity Scan since	Sep 06, 2016
Last build analyzed	a while ago
Language	C/C++
Repository URL	git://git.kernel.org/pub/scm/linux/kernel/git/next/linux-next
Homepage URL	https://git.kernel.org/cgit/linux/kernel/git/next/linux-next.git/
License	GPL (GNU General Public License version)

Want to view defects or help fix defects?

 Add me to project

Analysis Metrics

Version: next-20251022

Oct 22, 2025

34,355,053

0.55


Issues: By Snapshot | Outstanding Issues


Filters: Impact, Issue Kind, Classification, Component

CID	Type	Impact	Status	First...	Owner	Classification	Severity	Action	Component
100262	Dereferen...	Medium	New	02/24/06	Unassigned	Unclassified	Unspecified	Undecided	Sound...
100368	Explicit nu...	Medium	New	02/24/06	Unassigned	Unclassified	Unspecified	Undecided	Sound...
102446	Argument...	Medium	New	08/22/08	Unassigned	Unclassified	Unspecified	Undecided	Driver...
113840	Unchecke...	Medium	New	12/20/10	Unassigned	Unclassified	Unspecified	Undecided	Driver...

1307 issues match

<
Page
of 7
>





No Selection

Issues: By Snapshot | Outstanding Defects by category

DASHBOARDS

Quality Advisor
Security Advisor

ISSUES: BY SNAPSHOT

High Impact Outstanding
My Outstanding
Outstanding Issues
Outstanding Untriaged
All Newly Detected

Outstanding Defects by category

ISSUES: PROJECT SCOPE

All In Project

FILES

In Latest Snapshot

FUNCTIONS

High CCM (>15)
In Latest Snapshot

COMPONENTS

All In Project
High Issue Density (>1)
With Outstanding Issues

Issues: By Snapshot | Outstanding Defects by category



Filters: Status, Issue Kind, Classification

Category	# Items	CID	Type	Impact	Status	First...	Owner
Code maintainability...	278	1665977	Write to c...	High	New	09/20/25	Unassigned
Incorrect expression	368	1642386	Write to c...	High	New	02/03/25	Unassigned
Uninitialized variables	453	1641841	Write to c...	High	New	02/03/25	Unassigned
Data race undermin...	582	1538226	Write to c...	High	New	07/10/23	Unassigned

20 items match < Page 1 of 1 > 22 Issues match < Page 1 of 1 >





Coverity Scan is down

We're performing some upgrades to the Scan servers. During this time, login is disabled. All changes to projects and accounts are disabled.

We're really sorry for the inconvenience. We're working quickly to be back online soon!

Please follow us on twitter [@CoverityScan](https://twitter.com/CoverityScan) for updates.

Please contact scan-admin@coverity.com for any queries.

False Positive



```

1734 EXPORT_SYMBOL_GPL(vchiq_mmal_component_finalise);
1735
1736 /*
1737  * cause a mmal component to be destroyed
1738  */
1739 int vchiq_mmal_component_finalise(struct vchiq_mmal_instance *instance,
1740                                  struct vchiq_mmal_component *component)
1741 {
1742     int ret;
1743
1744     if (mutex_lock_interruptible(&instance->vchiq_mutex))
1745         return -EINTR;
1746
1747     if (component->enabled)
1748         CID 1408947: (#1 of 1): Unused value (UNUSED_VALUE)
1749         returned_value: Assigning value from disable_component(instance, component) to ret here,
1750         ret = disable_component(instance, component);
1751
1752     value_overwrite: Overwriting previous write to ret with value from destroy_component(instance, component);
1753     ret = destroy_component(instance, component);
1754     component->in_use = false;
1755     mutex_unlock(&instance->vchiq_mutex);
1756     return ret;
1757 }
1758 EXPORT_SYMBOL_GPL(vchiq_mmal_component_finalise);

```

```

/*
 * cause a mmal component to be destroyed
 */
int vchiq_mmal_component_finalise(struct vchiq_mmal_instance *instance,
                                  struct vchiq_mmal_component *component)
{
    int ret;

    if (mutex_lock_interruptible(&instance->vchiq_mutex))
        return -EINTR;

    if (component->enabled)
        ret = disable_component(instance, component);

    ret = destroy_component(instance, component);

    component->in_use = false;

    mutex_unlock(&instance->vchiq_mutex);

    return ret;
}
EXPORT_SYMBOL_GPL(vchiq_mmal_component_finalise);
90% drivers/platform/raspberrypi/vchiq-mmal/mmal-vchiq.c

```

[...]

Age	Commit message		Author	Files	Lines
13 hours	Merge tag 'powerpc-7.3-4' of git://git.kernel.org/pub/scm/linux/kernel/git/po...	HEAD master	Linus Torvalds	3	-2/+7
21 hours	powerpc/iommu: Fix the overflow validation in iommu_tce_check_ioba		Shivaprasad G Bhat	1	-1/+1
21 hours	KVM: PPC: Book3S HV: fix secure device page leak on uv_page_in() failure		Amit Machhiwal	1	-1/+4
21 hours	KVM: PPC: Book3S HV: fix use-after-free in kvmhv_emulate_tlbie_all_lpid()		Amit Machhiwal	1	-0/+2
34 hours	Merge tag 'sched_ext-for-7.3-rc3-fixes' of git://git.kernel.org/pub/scm/linux...		Linus Torvalds	7	-89/+254
34 hours	Merge tag 'cgroup-for-7.3-rc3-fixes' of git://git.kernel.org/pub/scm/linux/ke...		Linus Torvalds	1	-2/+5
36 hours	sched_ext: Maintain an online cid mask in the scheduler arena		Tejun Heo	4	-11/+78
36 hours	sched_ext: scx_qmap: Restore unused idle claims from ops.dispatch()		Tejun Heo	2	-4/+16
36 hours	Merge tag 'sysctl-7.03-fixes-rc4' of git://git.kernel.org/pub/scm/linux/kerne...		Linus Torvalds	1	-7/+9
46 hours	sysctl: Fix type truncation in sysctl_msec_to_jiffies		Joel Granados	1	-0/+2

[...]

Clone

[git://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git](https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git)

<https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git>

<https://kernel.googlesource.com/pub/scm/linux/kernel/git/torvalds/linux.git>

[**https://git.kernel.org/pub/scm/linux/kernel**](https://git.kernel.org/pub/scm/linux/kernel)

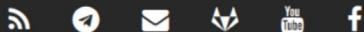


<https://flusp.ime.usp.br/kernel/Kernel-compilation-and-installation/>



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Kernel Compilation and Installation

Written by Rodrigo Siqueira

Written on February 16, 2019, last modified on February 26, 2025

In this tutorial, we will learn how to compile and install the Linux Kernel by using Linus Torvalds tree. We expect that you already have a virtual machine (preferably QEMU) available.

Simplified Patch Preparation

- git commit -s
- git format-patch -1 --base=v7.3-rc3 [--rfc=-RFT] [-v2] <commit>
- ./scripts/get_maintainer.pl <patch-file | source-code-file>
- git send-email [--dry-run] \
--to='m1@example.com, m2@example.com, r1@example.com' \
--cc='linux-kernel@vger.kernel.org, mailing-list@example.com' \
<patch-file>

<https://git-send-email.io/>



Documentation/process/submitting-patches.rst



```
$ ./scripts/get_maintainer.pl scripts/mod/modpost.c
Nathan Chancellor <nathan@kernel.org> (maintainer: KERNEL BUILD + files below scripts/ (unless mai...))
Nicolas Schier <nsc@kernel.org> (maintainer: KERNEL BUILD + files below scripts/ (unless mai...), commit_signer: 6/14=43%)
Sami Tolvanen <samitolvanen@google.com> (commit_signer: 2/14=14%)
Jani Nikula <jani.nikula@intel.com> (commit_signer: 2/14=14%, authored: 2/14=14%, added_lines: 58/110=53%, removed_lines: 58/72=81%)
Josh Poimboeuf <jpoimboe@kernel.org> (commit_signer: 2/14=14%, authored: 2/14=14%, added_lines: 6/110=5%)
Siddharth Nayyar <sidnayyar@google.com> (commit_signer: 2/14=14%, authored: 2/14=14%, added_lines: 12/110=11%, removed_lines: 4/72=6%)
Rong Xu <xur@google.com> (authored: 2/14=14%, added_lines: 13/110=12%, removed_lines: 4/72=6%)
Alexandre Courbot <acourbot@nvidia.com> (authored: 1/14=7%, added_lines: 10/110=9%)
linux-kbuild@vger.kernel.org (open list: KERNEL BUILD + files below scripts/ (unless mai...))
linux-kernel@vger.kernel.org (open list)
KERNEL BUILD + files below scripts/ (unless maintained elsewhere) status: Odd Fixes
```

_BlockWrite()

```
rtl8723b_hal_init.c
rtl8723b_hal_init.c /drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c Show File Path

/* 3 Phase #2 */
if (remainSize_p1) {
    CID 1408950: (#1 of 1): Unused value (UNUSED_VALUE)
    assigned_value: Assigning value from blockCount_p1 * blockSize_p1 to offset here, but that stored value is overwritten before it can be used.
    offset = blockCount_p1 * blockSize_p1;

    blockCount_p2 = remainSize_p1/blockSize_p2;
    remainSize_p2 = remainSize_p1%blockSize_p2;
}

/* 3 Phase #3 */
if (remainSize_p2) {
    value_overwrite: Overwriting previous write to offset with value from blockCount_p1 * blockSize_p1 + blockCount_p2 * blockSize_p2.
    offset = (blockCount_p1 * blockSize_p1) + (blockCount_p2 * blockSize_p2);

    blockCount_p3 = remainSize_p2 / blockSize_p3;

    for (i = 0; i < blockCount_p3; i++) {
        ret = rtw_write8(padapter, (FW_8723B_START_ADDRESS + offset + i), *(bufferPtr + offset + i));
    }
}
```

Check the Changes History

- git log \

-L :_BlockWrite:drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c \
a6f242ad1e2b

- git log \

-L 76,81:drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c \
a6f242ad1e2b

◆ On commit c1314fe4d28 (“staging: rtl8723bs: remove all RT_TRACE logs in hal/ and os_dep/”)

```
c1314fe4d28f staging: rtl8723bs: remove all RT_TRACE logs in hal/ and os_dep/
diff --git a/drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c b/drivers/staging/rtl8723bs/hal/rtl8723b_hal_init
.c
index 0273118e70af..b90b225d9589 100644
--- a/drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c
+++ b/drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c
@@ -90,21 +79,10 @@ static int _BlockWrite(struct adapter *padapter, void *buffer, u32 buffSize)
    if (remainSize_p1) {
        offset = blockCount_p1 * blockSize_p1;

        blockCount_p2 = remainSize_p1/blockSize_p2;
        remainSize_p2 = remainSize_p1%blockSize_p2;

        if (blockCount_p2) {
            RT_TRACE(
                _module_hal_init_c_,
                _drv_notice_,
                (
                    "_BlockWrite: [P2] buffSize_p2(%d) blockSize_p2(%d) blockCount_p
2(%d) remainSize_p2(%d)\n",
                    (buffSize - offset),
                    blockSize_p2,
                    blockCount_p2,
                    remainSize_p2
                )
            );
        }
    }
}
```



Subject: [PATCH] staging: rtl8723bs: remove unused offset in phase 2 _BlockWrite()

Date: Mon, 20 Apr 2026 11:46:42 +0700 [thread overview]

Message-ID: <20260420044651.164450-1-robertusdchris@gmail.com> (raw)

Commit c1314fe4d28f ('staging: rtl8723bs: remove all RT_TRACE logs in hal/ and os_dep/') removed the unnecessary RT_TRACE logs in hal/ and os_dep/ files, but left the variable "offset" in the Phase 2 _BlockWrite() function unused, so delete it.

This is reported by Coverity Scan with CID 1408950 as UNUSED_VALUE.

Signed-off-by: Robertus Diawan Chris <robertusdchris@gmail.com>

drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c | 2 --

1 file changed, 2 deletions(-)

diff --git a/drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c b/drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c

index e794fe3caf9d..988829d5f3c3 100644

--- a/drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c

+++ b/drivers/staging/rtl8723bs/hal/rtl8723b_hal_init.c

@@ -68,8 +68,6 @@ static int _BlockWrite(struct adapter *padapter, void *buffer, u32 buffSize)

/* 3 Phase #2 */

if (remainSize_p1) {

- offset = blockCount_p1 * blockSize_p1;

-

blockCount_p2 = remainSize_p1/blockSize_p2;

remainSize_p2 = remainSize_p1%blockSize_p2;

}

--

2.53.0



scarlett2_add_line_in_ctls()

```
6687 mixer_scarlett2.c /sound/usb/mixer_scarlett2.c Show File Path
6688
6689 /* Add software-controllable input gain controls */
6690 for (i = 0; i < info->gain_input_count; i++) {
6691     scnprintf(s, sizeof(s), fmt, i + 1,
6692             "Gain", "Volume");
6693     value_overwrite: Overwriting previous write to err with value from scarlett2_add_new_ctl(mixer, &scarlett2_input_gain_ctl, i, 1, s, &private->input_gain_ctls[i]).
6694     err = scarlett2_add_new_ctl(
6695         mixer, &scarlett2_input_gain_ctl,
6696         i, 1, s, &private->input_gain_ctls[i]);
6697     if (err < 0)
6698         return err;
6699     private->input_gain_ctls[i]->tlv.p =
6700         private->config_set->input_gain_tlv;
6701
6702     scnprintf(s, sizeof(s), fmt, i + 1,
6703             "Autogain", "Switch");
6704     err = scarlett2_add_new_ctl(
6705         mixer, &scarlett2_autogain_switch_ctl,
6706         i, 1, s, &private->autogain_ctls[i]);
6707     if (err < 0)
6708         return err;
6709
6710     scnprintf(s, sizeof(s), fmt, i + 1,
6711             "Autogain Status", "Enum");
6712     CID 1598781: (#1 of 1) Unused value (UNUSED_VALUE)
6713     returned_value: Assigning value from scarlett2_add_new_ctl(mixer, &scarlett2_autogain_status_ctl, i, 1, s, &private->autogain_status_ctls[i]) to err here, but that stored value is overwritten before it can be used.
6714     err = scarlett2_add_new_ctl(
6715         mixer, &scarlett2_autogain_status_ctl,
6716         i, 1, s, &private->autogain_status_ctls[i]);
6717 }
```

On commit 87b73d48a583 (“ALSA: scarlett2: Define the maximum preamp input gain per-config-set”)

```
/* Add software-controllable input gain controls */
for (i = 0; i < info->gain_input_count; i++) {
    scnprintf(s, sizeof(s), fmt, i + 1,
              "Gain", "Volume");

    err = scarlett2_add_new_ctl(
        mixer, &scarlett2_input_gain_ctl,
        i, 1, s, &private->input_gain_ctls[i]);
    if (err < 0)
        return err;
    private->input_gain_ctls[i]->tlv.p =
        private->config_set->input_gain_tlv;

    scnprintf(s, sizeof(s), fmt, i + 1,
              "Autogain", "Switch");
    err = scarlett2_add_new_ctl(
        mixer, &scarlett2_autogain_switch_ctl,
        i, 1, s, &private->autogain_ctls[i]);
    if (err < 0)
        return err;

    scnprintf(s, sizeof(s), fmt, i + 1,
              "Autogain Status", "Enum");
    err = scarlett2_add_new_ctl(
        mixer, &scarlett2_autogain_status_ctl,
        i, 1, s, &private->autogain_status_ctls[i]);
}

/* Add safe-mode input switch controls */
for (i = 0; i < info->safe_input_count; i++) {
```

70% sound/usb/mixer_scarlett2.c



git format-patch --base=v7.1-rc2 --rfc=-RFT ...

Subject: [PATCH **RFT**] ALSA: scarlett2: Add missing error check when initialise Autogain Status
Date: Fri, 8 May 2026 10:39:14 +0700 [thread overview]
Message-ID: <20260508033914.111596-1-robertusdchris@gmail.com> (raw)

When initialise new control with scarlett2_add_new_ctl() function for Autogain Status, scarlett2_add_new_ctl() might throw an error. So, add error check after initialise new control for Autogain Status.

This is reported by Coverity Scan with CID 1598781 as UNUSED_VALUE.

Fixes: 0a995e38dc44 ("ALSA: scarlett2: Add support for software-controllable input gain")
Signed-off-by: Robertus Diawan Chris <robertusdchris@gmail.com>

I don't have the device myself, so I can't test this change. Thank you.

```
sound/usb/mixer_scarlett2.c | 2 ++  
1 file changed, 2 insertions(+)
```

```
diff --git a/sound/usb/mixer_scarlett2.c b/sound/usb/mixer_scarlett2.c  
index 8eaa96222759..0f83f8981213 100644  
--- a/sound/usb/mixer_scarlett2.c  
+++ b/sound/usb/mixer_scarlett2.c  
@@ -6707,6 +6707,8 @@ static int scarlett2_add_line_in_ctls(struct usb_mixer_interface *mixer)  
    err = scarlett2_add_new_ctl(  
        mixer, &scarlett2_autogain_status_ctl,  
        i, 1, s, &private->autogain_status_ctls[i]);  
+  
+    if (err < 0)  
+        return err;  
+  
    }  
  
    /* Add autogain target controls */
```

base-commit: 7fd2df204f342fc17d1a0bfcd474b24232fb0f32

--
2.54.0

drivers/staging/fbtft/README

FBTFT

Linux Framebuffer drivers for small TFT LCD display modules.
The module 'fbtft' makes writing drivers for some of these displays very easy.
Development is done on a Raspberry Pi running the Raspbian "wheezy" distribution.
For new hardware support consider using DRM subsystem (see TODO).

NOTE:

The driver is in maintenance mode, only performance issue or bug fixes are accepted, which effectively means the patches must be tested on the real hardware (the patch must be accompanied with the information what hardware is that). The treewide changes may also be accepted as an exception.

Running tests with kunit_tool

We can either run KUnit tests using `kunit_tool` or can run tests manually, and then use `kunit_tool` to parse the results. To run tests manually, see: [Run Tests without kunit_tool](#). As long as we can build the kernel, we can run KUnit.

`kunit_tool` is a Python script which configures and builds a kernel, runs tests, and formats the test results.

Run command:

```
./tools/testing/kunit/kunit.py run
```

We should see the following:

```
Configuring KUnit Kernel ...  
Building KUnit kernel...  
Starting KUnit kernel...
```

We may want to use the following options:

```
./tools/testing/kunit/kunit.py run --timeout=30 --jobs=`npr
```

- `--timeout` sets a maximum amount of time for tests to run.
- `--jobs` sets the number of threads to build the kernel.

Linux Kernel Selftests

The kernel contains a set of “self tests” under the `tools/testing/selftests/` directory. These are intended to be small tests to exercise individual code paths in the kernel. Tests are intended to be run after building, installing and booting a kernel.

Kselftest from mainline can be run on older stable kernels. Running tests from mainline offers the best coverage. Several test rings run mainline kselftest suite on stable releases. The reason is that when a new test gets added to test existing code to regression test a bug, we should be able to run that test on an older kernel. Hence, it is important to keep code that can still test an older kernel and make sure it skips the test gracefully on newer releases.

On some systems, hot-plug tests could hang forever waiting for cpu and memory to be ready to be offlined. A special hot-plug target is created to run the full range of hot-plug tests. In default mode, hot-plug tests run in safe mode with a limited scope. In limited mode, `cpu-hotplug` test is run on a single cpu as opposed to all hotplug capable cpus, and memory hotplug test is run on 2% of hotplug capable memory instead of 10%.

kselftest runs as a userspace process. Tests that can be written/run in userspace may wish to use the [Test Harness](#). Tests that need to be run in kernel space may wish to use a [Test Module](#).



qcomtee_object_user_init()

```
core.c
core.c /drivers/tee/qcomtee/core.c

299
300
1. va_init: Initializing va_list ap.
301 va_start(ap, fmt);
2. Switch case value QCOMTEE_OBJECT_TYPE_CB.
302 switch (ot) {
303 case QCOMTEE_OBJECT_TYPE_NULL:
304     ret = 0;
305
306     break;
307 case QCOMTEE_OBJECT_TYPE_CB:
308     object->ops = ops;
3. Condition !object->ops->dispatch, taking true branch.
309 if (!object->ops->dispatch)
    CID 1666903: (#1 of 1): Missing varargs init or cleanup (VARARGS)
    4. missing_va_end: va_end was not called for ap.
310     return -EINVAL;
311
312     /* If failed, "no-name". */
313     object->name = kvasprintf_const(GFP_KERNEL, fmt, ap);
314     QCOMTEE_OBJECT_SET(object, QCOMTEE_OBJECT_TYPE_CB);
315
316     ret = 0;
317     break;
```

va_end

Defined in header `<stdarg.h>`

```
void va_end( va_list ap );
```

The `va_end` macro performs cleanup for an `ap` object initialized by a call to `va_start` or `va_copy`. `va_end` may modify `ap` so that it is no longer usable.

If there is no corresponding call to `va_start` or `va_copy`, or if `va_end` is not called before a function that calls `va_start` or `va_copy` returns, the behavior is undefined.



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qcomtee_object_user_init() is a variadic function and when the function return because there's no dispatch callback in QCOMTEE_OBJECT_TYPE_CB case, there's no va_end to cleanup "ap" object initialized by va_start and that can cause undefined behavior. So make sure to use va_end before returning the error code when there's no dispatch callback.

This is reported by Coverity Scan as "Missing varargs init or cleanup".

Fixes: d6e290837e50 ("tee: add Qualcomm TEE driver")

Signed-off-by: Robertus Diawan Chris <robertusdchris@gmail.com>

I don't have the device, so I am not sure how to test this change.

Thank you.

```
drivers/tee/qcomtee/core.c | 8 +++++--
1 file changed, 6 insertions(+), 2 deletions(-)

diff --git a/drivers/tee/qcomtee/core.c b/drivers/tee/qcomtee/core.c
index b1cb50e434f0..901a31e8201f 100644
--- a/drivers/tee/qcomtee/core.c
+++ b/drivers/tee/qcomtee/core.c
@@ -306,8 +306,10 @@ int qcomtee_object_user_init(struct qcomtee_object *object,
     break;
     case QCOMTEE_OBJECT_TYPE_CB:
         object->ops = ops;
-        if (!object->ops->dispatch)
-            return -EINVAL;
+        if (!object->ops->dispatch) {
+            ret = -EINVAL;
+            goto out;
+        }

         /* If failed, "no-name". */
         object->name = kvasprintf_const(GFP_KERNEL, fmt, ap);
@@ -320,6 +322,8 @@ int qcomtee_object_user_init(struct qcomtee_object *object,
     default:
         ret = -EINVAL;
     }
+
+out:
     va_end(ap);

     return ret;

base-commit: 5d6919055dec134de3c40167a490f33c74c12581
--
2.54.0
```

Hi,

Forgot to mention: how about using `a break instead of a goto.`
Then feel free to add Reviewed-by.

Thanks,
Amir

Send Next Version of Patch

- Send the next version of the patch as new email
- Put the changelogs in the next version patch
- Wait a few days (or even weeks) before sending next version of patch

Scenario Send Next Version of Patch Too Fast

- Maintainer 1 give feedback to the patch (version 1's thread)
- Send version 2 immediately
- Maintainer 2 didn't agree with Maintainer 1's approach, and give different approach (version 2's thread)
- Send version 3 immediately
- Maintainer 1 pointed out the downside of Maintainer 2's approach (version 3's thread)



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qcomtee_object_user_init() is a variadic function and when the function return because there's no dispatch callback in QCOMTEE_OBJECT_TYPE_CB case, there's no va_end to cleanup "ap" object initialized by va_start and that can cause undefined behavior. So make sure to use va_end before returning the error code when there's no dispatch callback.

This is reported by Coverity Scan as "Missing varargs init or cleanup".

Fixes: d6e290837e50 ("tee: add Qualcomm TEE driver")

Signed-off-by: Robertus Diawan Chris <robertusdchris@gmail.com>

Reviewed-by: Amirreza Zarrabi <amirreza.zarrabi@oss.qualcomm.com>

v1 -> v2:

- Use "break" statement instead of "goto" statement. There's va_end outside of switch-case, so we only need to go out from switch-case instead of using a label (suggested by Amirreza Zarrabi).
- Add "Reviewed-by" tag from Amirreza Zarrabi.

v1:

<https://lore.kernel.org/all/20260513091031.145826-1-robertusdchris@gmail.com/>

I don't have the device, so I am not sure how to test this change.
Thank you.

drivers/tee/qcomtee/core.c | 6 +++--

1 file changed, 4 insertions(+), 2 deletions(-)

diff --git a/drivers/tee/qcomtee/core.c b/drivers/tee/qcomtee/core.c

index b1cb50e434f0..60fe3b5776e3 100644

--- a/drivers/tee/qcomtee/core.c

+++ b/drivers/tee/qcomtee/core.c

```
@@ -306,8 +306,10 @@ int qcomtee_object_user_init(struct qcomtee_object *object,
    break;
    case QCOMTEE_OBJECT_TYPE_CB:
        object->ops = ops;
-       if (!object->ops->dispatch)
-           return -EINVAL;
+       if (!object->ops->dispatch) {
+           ret = -EINVAL;
+           break;
+       }

    /* If failed, "no-name". */
    object->name = kvasprintf_const(GFP_KERNEL, fmt, ap);
```

base-commit: 5200f5f493f79f14bbdc349e402a40dfb32f23c8

--

2.54.0



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Bluetooth: btintel_pcie: Remove unused ret in btintel_pcie_submit_rx_work()

Message ID 20260430040518.163212-1-robertusdchris@gmail.com ([mailing list archive](#))
State New, **archived**
Submitter Robertus Diawan Chris
Series 1 of 1
Date April 30, 2026, 4:05 a.m.

https://patchwork.linuxtv.org/project/linux-media/list/

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16
17
>

Patch	Series	A/C/F/R/T	S/W/F	▲ Date	Submitter	Delegate	State
[v2] media: dvb-frontends: ds3000: validate firmware size	[v2] media: dvb-frontends: ds3000: validate firmware size	-- 1 --	-- --	2026-09-20	Pengpeng Hou		New
[v2] media: cec: extron: clear response tail before dispatch	[v2] media: cec: extron: clear response tail before dispatch	-- 1 --	-- --	2026-09-20	Pengpeng Hou	hverkuil	New
media: dvb_ca_en50221: bail out of write loop on non-positive fraglen	media: dvb_ca_en50221: bail out of write loop on non-positive fraglen	-----	-- --	2026-09-19	Muhammad Bilal		New
media: v4l2-subdev: fix NULL deref in subdev_open() racing with unbind	media: v4l2-subdev: fix NULL deref in subdev_open() racing with unbind	- 1 1 --	①	2026-09-19	Nguyen Ngoc Thang		New
[next] media: i2c: os02g10: Fix spelling mistake "foramt" -> "format"	[next] media: i2c: os02g10: Fix spelling mistake "foramt" -> "format"	-----	-- --	2026-09-19	Colin Ian King		New
media: rkvddec: Remove redundant dev_err()	media: rkvddec: Remove redundant dev_err()	-----	-- --	2026-09-19	Diederik de Haas		New
[v3] media: ipu-bridge: Add OV13858 sensor config	[v3] media: ipu-bridge: Add OV13858 sensor config	---- 2	-- --	2026-09-19	Sergey Lebedev		New
[4/4] arm64: dts: qcom: nord-rrd: Enable iris video codec node	media: iris: Add support for nord platform	-----	-- --	2026-09-19	Vishnu Reddy		New

**“The beginning is not for proving
yourself, it is for learning the craft.”**

Thank You



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